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EDUCATION

- **Harvard University** Boston, MA
M.S. in Health Data Science *August 2025 – March 2027*
 - Program requirements complete December 2026 (commencement March 2027); available for full-time start January 2027
 - Coursework: Deep Learning, AI in Medicine, Healthcare Machine Learning, Data Science
- **Emory University, College of Arts and Sciences** Atlanta, GA
B.S. in Applied Mathematics and Statistics; Minor in Computer Informatics; GPA: 3.925 *August 2021 – May 2025*
 - Coursework: Probability and Statistics, Machine Learning, NLP, Data Structures, Database Systems

TECHNICAL SKILLS

- **Languages:** Python, R
- **ML:** PyTorch, scanpy, AnnData, RAG, OpenAI
- **Backend:** Flask, REST APIs, React, Auth0, MongoDB
- **Cloud:** AWS Elastic Beanstalk, CloudFront, Docker
- **Certifications:** AWS Certified Cloud Practitioner, AWS Certified AI Practitioner, AWS Machine Learning Associate

CLINICAL AI EXPERIENCE

- **Airway Management Simulation Chatbot** Atlanta, GA
Full-Stack Engineer, Scrum Master — Emory Pediatric Hospital *January 2024 – August 2025*
 - Built a HIPAA-conscious clinical simulation platform with a Flask REST API, React, MongoDB, and Auth0, deployed on AWS Elastic Beanstalk and CloudFront.
 - Retrieval runs on MongoDB Atlas Vector Search in the same datastore as application persistence.
 - Worked on an Agile team of 6 with code reviews and a clinician feedback loop.

MACHINE LEARNING PROJECTS

- **speciesOT: Cross-Species Single-Cell Translation** Boston, MA
Graduate Researcher — Mooney Lab & Alvarez-Melis Lab, Wyss Institute *February 2026 – Present*
 - Built a mouse-to-human single-cell translation that compares an ICNN optimal-transport map with an scGen latent mean-shift.
 - On a paired held-out-species evaluation, scGen R^2 was **0.85–0.90** with the source species in training versus **0.65–0.67** when that species was held out.
 - Gene-space ranking reversed with the split (6619 orthologs vs 1000 HVGs), so the biologically convenient gene set is not automatically the better predictor.
- **mixhvg-py: Validated Python Port of an R HVG Ensemble** Boston, MA
Sole author *July 2026*
 - Ported mixhvg, an ensemble highly variable gene selector, to an AnnData-native API used by speciesOT specs. Zhao et al. own the method.
 - Validated against R / scan / Seurat. Seurat methods match at Spearman **1.0000**. The paper-recommended mixture agrees at Jaccard **0.980**.
 - A weighted-mean log-space correction overshot scan and raised discarded genes from 0.499 to 0.719. A weighted median restored 0.503 vs R 0.499.